

ECONOMETRICS _ PROJECT 2

You are given a dataset of 19253 households in 14 provinces of the Central Coast Region of Vietnam

- **Explanatory variables**
 - age: age of the household head (years)
 - edu: the number of schooling years of the household head
 - gender of the household head: 1 = male; 0 = female
 - married: marital status of the household head: "1=married; 0=otherwise"
 - ethnicity: ethnicity of the household head: 1= Kinh; 0 = ethnic minority
 - hhsize: total household members
 - dep_ratio: the dependency ratio is calculated by dividing the number of dependents by the household size.
 - aland: the size of annual cropland (m²)
 - pland: the size of perennial cropland (m²)
 - fland: the size of forestland (m²)
 - gland: the size of garden land (m²)
 - province: a categorical variable including 14 categories (name of provinces).
 - urban: 1=living in urban areas; 0=living in rural areas
 - year: 1=2018; 0=2016
- **The dependent variable (income):** household income per capita/month (thousand VND)

The requirements of the exercise are as follows

1. Specify an economic model and an econometric model showing the factors that affect household income per capita. Discuss the reasons for selecting explanatory variables and what their expected signs are.
2. Discuss which assumptions of multiple linear regression (MLR) need to be satisfied to obtain unbiased coefficient estimates from the OLS estimator for the model. Discuss each assumption in detail in the context of the model.

Take the natural log of land variables and then regress income on all other explanatory variables. Please test for homoscedasticity and multicollinearity. If the variance of the error is not constant, how do you deal with this issue?

3. Holding all other factors in the model fixed, what is the difference in income levels between the following four groups?

Group 1: urban and male-headed households

Group 2: urban and female-headed households

Group 3: rural and male-headed households

Group 4: rural and female-headed households (base group)

What is the income difference between the base province (Thanh Hoa) and other provinces?

What is the income difference between the two years?

4. Test the null hypothesis that the effect of education on income is the same between Kinh households and ethnic minority households. Test the hypothesis that the household income per capita is 17% higher in 2018 than in 2016.
5. Use Stata to draw a graph showing how education has different effects (different slopes) on income between Kinh households and ethnic minority households.
Use `outreg2` to get the regression outputs and then present the results professionally.
6. Please summarize the main findings and propose policy recommendations based on the findings.

Note: You should take log (X+1) for land and income variables.

Words count: maximum 3000 words, excluded references

Standard format for the main document:

- 1) Size A-4 A4 (210x297 mm) or United States letter size (8.5x11 inches) with 2.5 cm margins all around. If possible, avoid footnotes and appendices.
- 2) Use Times New Roman typeface with a 12-pitch font (12 characters per inch) for the body of the text and the tables.
- 3) The heading Introduction is not needed unless it contains additional text. Use only three levels of headings: Level 1, 12 pt, bold, lower case except for first letter in the first word; Level 2, 12 pt, bold/italics, lower case as above; Level 3, 12 pt italics, lower case as above.
- 4) The text should be double-spaced and left-justified.
- 5) Indent all paragraphs except the first one under a heading; right-hand margin is unjustified; no extra space between paragraphs.
- 6) Tables/figures are to be submitted as separate files with self-explanatory legends. Indicate the approximate location for tables and figures in the text, e.g. Table 1 to feature here. Figures are to be submitted as Camera Ready Copy, that is, ready for printing, using an optional typeface that is clear and easy to read. Make sure the figures are legible even when reduced in size for publication.